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Percutaneous CT-Guided Cryoablation of the Bilateral Pudendal Nerves for Palliation of Intractable Pain Related to Pelvic Neoplasms

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Abstract

Objectives: The purpose of this report is to describe the effect of computed tomography–guided bilateral pudendal nerve cryoablations on pain and time to discharge in the setting of acute hospitalizations secondary to refractory pelvic pain from cancer. **Methods:** Investigators queried the medical record for patients who underwent pudendal nerve cryoablation using the Category III Current Procedural Technology code assignment 0442T or Category I code 64640 for cases prior to 2015. The resulting list was reviewed, and procedures performed on inpatients for intractable pelvic pain related to neoplasm were selected. The final cohort was then analyzed with regard to patient demographics, procedure details, technical success, safety, pain scores, and time to discharge. **Results:** Ten patients underwent cryoablation by 3 operators for palliation of painful pelvic neoplasms between June 2014 and January 2019. All probes were satisfactorily positioned and freeze cycles undertaken without difficulty. There were no procedure-related complications or adverse events. The mean difference in pre- and posttreatment worst pain scores was significant (n = 5.20, $P = .003$). The mean time to discharge following the procedure was 2.3 days. **Conclusion:** Computed tomography–guided percutaneous cryoablation of the bilateral pudendal nerves may represent a viable option in the setting of acute hospitalization secondary to intractable pain in patients with pelvic neoplasms.

Keywords

cryoablation, percutaneous, palliative, pelvic neoplasm, pudendal nerves, interventional radiology

Introduction

Hospitalization for intractable pain is a common clinical scenario for patients with locally advanced neoplastic disease in the pelvis. Palliative radiation therapy may not be an option or may have been exhausted, direct ablation of the mass is often not feasible, and most of these patients are not surgical candidates—creating a formidable barrier to timely palliation and/or discharge.[1,2]

Percutaneous image-guided cryoneurolysis is evolving as an effective treatment option for several historically difficult to manage pain syndromes, including those related to cancer.[3-8] The purpose of this report is to describe effect of a novel procedure involving computed tomography (CT)–guided bilateral pudendal nerve cryoablations on pain and time to discharge in the setting of acute hospitalizations secondary to refractory pelvic pain from cancer.

Methods

This retrospective review was approved by the local institutional review board. All activities related to collected patient data complied with the Health Insurance Portability and Accountability Act (decision/protocol number: 00078244). Investigators queried the medical record for patients who underwent pudendal nerve cryoablation using the Category III Current Procedural Technology code assignment 0442T or Category I code 64640 for cases prior to 2015. The resulting cohort was reviewed, and procedures performed on inpatients for intractable pelvic pain related to neoplasm were selected. Patient demographics are included in Table 1.

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Table 1. Patient Demographics.

Age/sex	Primary lesion	Presenting symptoms	Time to discharge following procedure
54/F	Rectal mass, primary rectal small cell neuroendocrine tumor	Admitted to inpatient service for intractable, burning, sharp pelvic and rectal pain not responding to medical regimen	2 days
32/F	Rectal mass, HIV lymphoma, rectovaginal fistula	Admitted with intractable rectal and vaginal pain, unresponsive to narcotics	2 days
51/F	Rectosigmoid mass, primary colon cancer	Admitted to inpatient service for intractable, pelvic and rectal pain not responding to medical regimen	Same day
47/F	Anal cancer, squamous cell primary	Admitted to inpatient service for intractable, pain related to a perineal anal-cutaneous fistula	1 day
73/F	Vaginal carcinoma, squamous cell primary	Admitted to inpatient service with unmanageable, intractable pelvic and vaginal pain refractory to medical regimen	Same day
78/F	Bladder cancer, undifferentiated urothelial cell origin	Admitted to inpatient service with intractable, pelvic pain related to bladder mass	10 days (multiple medical problems)
63/F	Bladder cancer, urothelial carcinoma primary	Admitted to inpatient service with multiple problems, including obstructive uropathy and pain	2 days
71/F	Colorectal cancer	Admitted to inpatient service with intractable anal and rectal pain	5 days
89/F	Cervical cancer	Admitted to inpatient service with intractable pelvic pain and vulvodynia	Same day
56/M	Rectal CA	Admitted with intractable rectal/anal pain s/p radiation therapy	1 day

Pain was considered appropriate for this procedure if it involved the distribution of the pudendal nerve—namely the genitals, perineum, and/or rectum.[9] Each patient underwent CT-guided percutaneous cryoablation of the bilateral pudendal nerves with a 17-gauge cryoablation probe (IceSphere, Galil/BTG). Using CT guidance, a percutaneous puncture site was marked such that a cryoablation probe could be safely guided to a position parallel and medial to the pudendal nerve as it courses proximally in the pudendal canal (Figure 1).[10] For each case, the freeze cycle was 8 minutes long, followed by a 4-minute passive thaw, a second 8-minute freeze, and a final 4-minute passive thaw—with the intention of including the pudendal nerve in a −20 °C to 40 °C zone of decreased temperature without contacting the nerve directly with the needle. Procedures were performed under moderate sedation with midazolam (Hospira) and fentanyl (West-Ward Pharmaceuticals).

Technical success was defined as successful placement of the cryoablation probe percutaneously, using CT guidance, such that the bilateral proximal pudendal nerves were included in the predicted ablation zones. In addition, concluding technical success for this procedure required that no procedure-related adverse events (AEs) had occurred.

Safety was quantified by the rate of procedure-related events (AEs occurring within 24 hours following the procedure), breakthrough events (AEs occurring at any time that required emergency or urgent physician consultation), AEs, and/or serious AEs that occurred during the patient's hospitalization or documented follow-up.

At the authors' institution, pain is logged in the medical record as part of scheduled routine vital sign measurements, every 6 hours. The highest (worst pain) score on a visual analog scale (VAS) was recorded during the 24 hours prior to and following the procedure.[11] Time to discharge following the procedure was calculated using documentation in the electronic medical record.

Descriptive analysis of the data was performed, as well as tests for normality (Shapiro-Wilk). Shapiro-Wilk is a test for normality of sample data. The results of a test for normality are used to determine whether parametric (if normally distributed) or nonparametric (if not normally distributed) statistical comparison testing is used. Means comparison was performed with a test of 3 VAS points difference in worst pain score pre- and posttreatment using a 1-sample t test.[11-13] SPSS v24 (IBM Corp) was used for all statistical analysis.

Results

Ten patients underwent cryoablation by 3 operators for palliation of painful pelvic neoplasms between June 2014 and January 2019. All probes were satisfactorily positioned and freeze cycles undertaken without difficulty. There were no procedure-related complications or AEs.

The difference between pre- and posttreatment pain scores was non-normally distributed ($P = .099$). The mean difference in pre- and posttreatment worst pain scores was significant ($n = 5.20$, $P = .003$). The mean time to discharge following the procedure was 2.3 days.

Five patients were seen in interventional radiology clinic following discharge (average time to follow-up = 12.8 ± 3.7 days). No additional AEs or complications were uncovered. Of these, 4 had recorded VAS scores (mean 2.5 ± 2.1). An additional patient was seen in palliative care clinic 10 days after discharge and recorded a pain score of 10/10 which was also associated with developments of several other unrelated pain sites. Three patients were discharged to hospice, and 1 patient died 28 days after discharge without other follow-up.

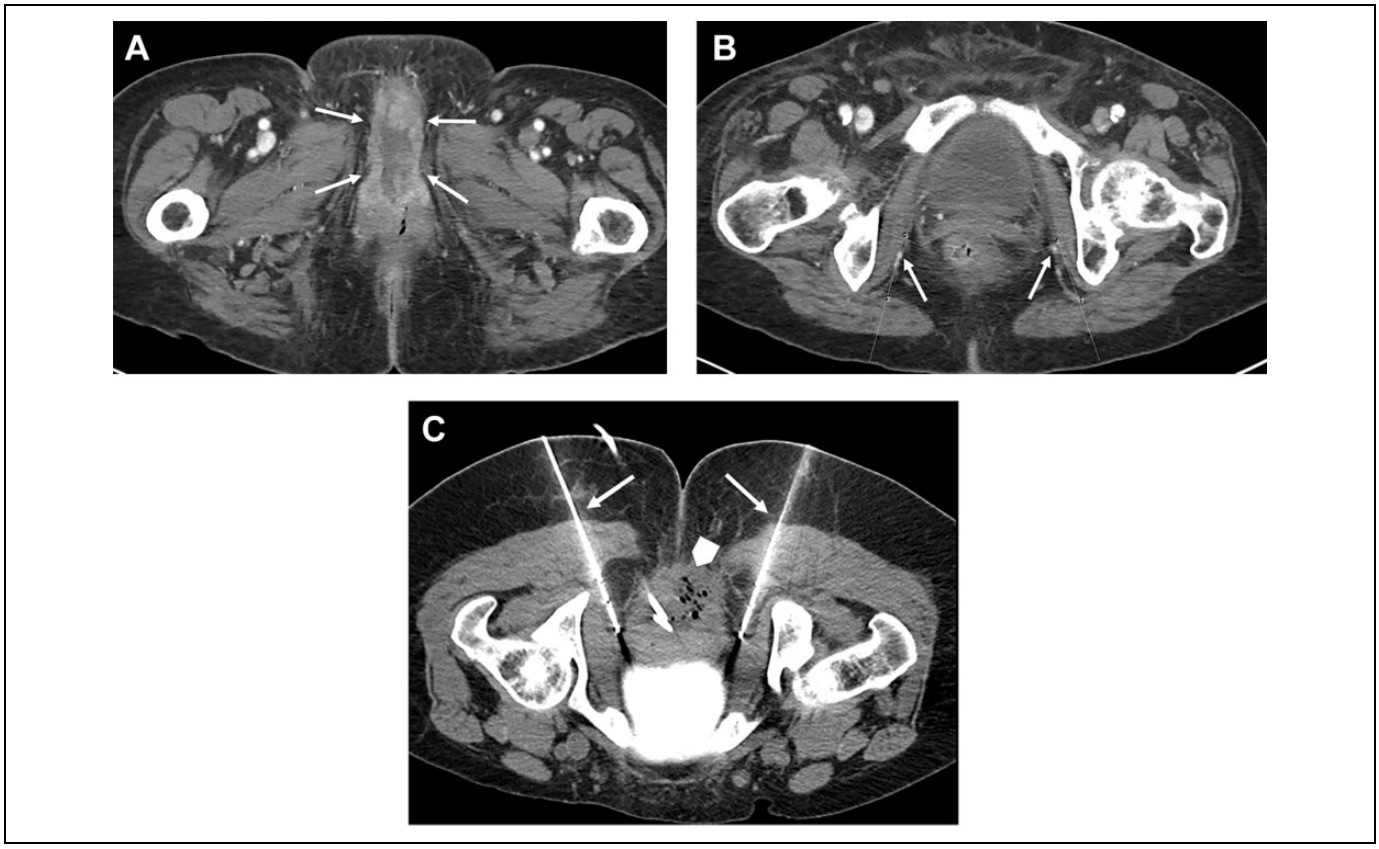


Figure 1. Representative images from pudendal nerve cryoablation procedures performed for palliation of pain related to neoplastic pelvic disease. A, Axial contrast-enhanced computed tomography (CT) image from a 73-year-old female with vaginal carcinoma demonstrates abnormal enhancement and soft tissue in the distribution of the pudendal nerves—vagina and perineum in this case. B, Separate image from same scan demonstrating planning markup for cryoablation procedure. The pudendal arteries can be seen as surrogate markers of the canals in this case (arrows). C, Axial CT intraprocedural image from a 54-year-old female with rectal cancer demonstrating cryoablation probes in the pudendal canals bilaterally. A necrotic rectal mass (arrowhead) and percutaneously placed drainage catheter are also visible.

Discussion

Approximately 10% of patients with cancer-related pain have symptoms that are difficult to manage with oral or parenteral analgesic medications ("intractable pain"). For these patients, consideration of invasive techniques for pain control is considered appropriate.[14]

The use of cryoablation to target nerves for the treatment of painful conditions has existed for more than 50 years.[15] Percutaneous cryoneurolysis has been reported in the settings of occipital neuralgia, phantom limb pain, muscle spasticity, knee pain, postoperative pain (shoulders, lower extremities, intercostal), peripheral neuropathy, inguinodynia, cancer pain, and many more.[3-8,16-23]

The integration of advanced imaging guidance to the management of pain provides unique opportunities for palliation in the setting of neoplastic disease by allowing deep structures and nerves to be safely accessed and ablated percutaneously.[16] Computed tomography guidance affords additional available targeting when compared with ultrasound by negating ultrasound degrading artifacts caused by deep tissue locations, adjacent air, or bone. Cold is applied to the nerves using closed end needle systems. The mechanism of effect that follows is a reproducible axonal injury and microtubule dissolution, subsequent Wallerian degeneration, and slow neuronal regeneration guided by intact connective tissue (vs cell death induced by osmotic gradient shifts when targeting tumors). The cumulative end point of these interventions is decreased pain sensation resulting from conduction attenuation, activation of descending inhibition, blockade of excitatory transmitter systems, and a generalized decrease in sodium channel activity.[5,17]

The pudendal nerve carries sensation from the genitals, perineum, distal portion of the rectum, and skin around the anus. Anatomically it enters the pudendal canal below the sacrotuberous ligament and continues anteroinferiorly along the lateral border of the ischioanal fossa, superficial to the obturator internus muscle with the internal pudendal artery and vein.[9,10] The pudendal nerve also contains motor fibers that contribute—along with the sacral plexus, lumbar plexus, and several other peripheral nerves—to the external anal and urethral sphincter muscles and some muscles of the pelvic floor.[18] The pudendal nerve has long been a target for intervention (block, ablation,

modulation, surgical) for the treatment of pelvic pain syndromes, including reports describing efficacy of cryoablation, radiofrequency ablation, and pulsed neuromodulation in the setting of nonneoplastic disease.[19-22]

This report reflects our initial clinical experience with bilateral CT-guided pudendal nerve cryoablation in the setting of hospitalization secondary to intractable pain from a pelvic neoplasm. Following cryoablation of the bilateral pudendal nerves, patients reported an average worst pain score decrease of 5.2 in 24 hours, and the average time to discharge was 2.3 days following the procedure—significant findings for patients who undergo interventions in the setting of chronic pain.[11-13,23] There were no procedure-related complications or AEs. That said, the procedure itself embodies the risks associated with percutaneous image-guided procedures such as bleeding and infection. Contraindications to implementation of this procedure to patients care include those accepted for traditional percutaneous procedures including bleeding diatheses, inability to consent, and/or unusual anatomy precluding safe access to targeted structures. The nature of the patient population precludes detailed follow-up, as they have advanced neoplastic disease and are largely comfort care.

Of note, no preprocedure test injection was performed—which may have attenuated the potential response rate to the intervention by including patients who may not have definitely responded to the block. A patient's response to treatment is dependent on the correct identification of the nerve responsible for the pain symptoms and subsequently, successful treatment of that nerve. Given the subjective nature of pain ratings and the potential for false-positive responses, it is often useful to combine CT imaging with a comparative block approach (using a diagnostic block to select patients for cryoablation) to augment efforts of both target identification and therapeutic treatment.[24] This 2-step procedure allows true pain-generating targets to be identified because, when successfully blocked, the attenuated target should consistently yield substantial, if not complete, relief in pain symptoms.[25] In the context of the current procedures, though, the 2-step procedure was foregone in order to minimize the number of procedures and delay to relief associated with multiple interventions.

Interpretation of these results is further limited by the small size of the cohort and retrospective nature of the study. Moreover, there is no long-term follow-up that can be used to assess the durability of the procedure. What may be useful in this population, though, is that in the setting of an acute hospitalization related to pain from pelvic neoplasm—when patients have exhausted other options—cryoablation of the bilateral pudendal nerves is a feasible and safe option.

With regard to cost, the bilateral CT-guided pudendal nerve cryoablation procedure in the setting of pelvic neoplasm may be considered an orphan or humanitarian procedure. At the same time, Category III codes were assigned to these procedures in 2015, acknowledging the potential for future conversion to Category I given the expanding indications for these techniques.[26-28] Clinically, the cases described herein represent clinical scenarios for which no other reasonable therapeutic option remained. As a result, a contextual discussion of other available treatments for patients in this clinical scenario does not apply.

Computed tomography–guided percutaneous cryoablation of the bilateral pudendal nerves may represent a viable option in the setting of acute hospitalization secondary to intractable pain in patients with pelvic neoplasms. Future prospective studies addressing length of stay, opioid requirements, and cost will further define the role of these procedures in this setting.

Authors' Note

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Declaration of Conflicting Interests

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